Online Supplementary Material

A Synergy of Quantum-Inspired Pathfinding Algorithms, Reinforcement Learning, and Graph Neural Networks

Amir Hossein Shahriari, Master of Research thesis, School of Computing, Macquarie University, 2026

This document holds the diagnostic, superseded and index tables that the thesis cites as Supplement S.1–S.5 and Tables S1–S11. Section, table, figure and equation numbers without an S refer to the thesis. Artefact paths are relative to `https://github.com/Amir-Shahriari/Thesis_Project`.

S.1 Training-Budget Tiers and the Headline Scale

Metric (warm V1)	1× **(10 it)**	4× **(40 it)**
Strict goal-reach rate	22 / 25 (88 %)	**25 / 25 (100 %)**
Strict-win rate (warm < cold)	21 / 25 (84 %)	**24 / 25 (96 %)**
Mean cost ratio vs Dijkstra	17.3×	14.8×
Median cost ratio vs Dijkstra	10.5×	4.3×
Cells at ≤ 5× Dijkstra	—	13 / 25
Cold strict goal-reach (reference)	3 / 25 (12 %)	2 / 25 (8 %)

Table S1: 50 × 50 V1 warm results, 1× versus 4× tier, 5 seeds × 5 scenarios, batch 64. Training longer advances the perturbation schedule, so every across-tier comparison in this section is made on different realisations of each graph. Unmatched: all 22 comparable cells have a different Dijkstra reference, and the three cells unsolvable at 1× (`runs/eval_reachability_audit.json`) are connected at 4×. Reach supports no compute claim (three discordant against zero, McNemar $p = 0.25$); cost ratios are indicative only. Sources: `runs/sweep_50x50/aggregate_50x50.json`, `runs/sweep_50x50/sweep_v1_50x50_4x.json`.

Metric	**V1 (warm, 100×100)**	**V1 (warm, 25×25, from §4.3.1)**
Strict goal-reach rate	**6 / 25 (24 %)**	24 / 25 (96 %)
Reasonable-reach $\mathcal{R}_3$	0 / 25 (0 %)	36 %
Mean composite cost (reached)	2,248 ± 1,187	242 ± 197
Mean cost ratio vs Dijkstra	40.11	4.42
Mean per-trajectory wall-clock (ms)	99.4 ± 33.2	7.5 ± 10.5
Strict-win rate (warm < cold)	24 %	84 %
Paired t-test p-value	untestable[§]	0.0074

Table S2: V1 at 100 × 100 against the 25 × 25 baseline. Three cells are unsolvable at their evaluation states; over solvable cells warm reach is 6/22 (27.3 %). [§]Untestable: warm reaches 6 cells and cold 1, disjoint, so there are no pairs. The reach contrast is 6 discordant cells against 1, exact McNemar $p = 0.125$ (§4.3.3). Source: `runs/sweep_v1_on_100x100.json`.

Seed	Scenario id	1× reach	4× reach	4× cost	Cost ratio	Train (min)
42	seed191664964_s0	No	Yes	1,035	15.73	137.4
42	seed191664964_s1	No	No	—	—	35.1
42	seed191664964_s2	No	Yes	1,221	23.29	59.6
42	seed191664964_s3	No	No	—	—	58.9
1337	seed1173222464_s0	No	Yes	3,556	60.06	35.0
1337	seed1173222464_s1	No	Yes	3,530	59.83	58.9
1337	seed1173222464_s2	No	No	—	—	58.9
2024	seed518677876_s1	No	Yes	4,638	82.88	35.1
2024	seed518677876_s2	No	No	—	—	81.5
2024	seed518677876_s4	No	Yes	776	15.22	181.4
7	seed2029167941_s2	No	Yes	3,200	59.25	36.4
7	seed2029167941_s3	No	Yes	579	9.08	124.2
7	seed2029167941_s4	No	No	—	—	144.3
314159	seed1830621171_s0	No	Yes	4,662	84.26	35.0
314159	seed1830621171_s2	No	Yes	1,933	28.76	145.6
Aggregate		0 / 15 (0 %)	**10 / 15 (66.7 %)**	—	mean 43.84×	mean 81.8
95 % binomial CI on 4× reach rate		[0.384, 0.882]				
McNemar's test (1× vs 4×)		$p = 0.0020$				

Table S3: V1 long-train subset: 15 cells spanning all five seeds (two to four per seed), drawn from cells where the 1× budget failed; evaluated at iterations 10 and 40. The 4× instances are easier: 11 of 12 cells with a reference at both have a shorter optimum (sign test $p = 0.0063$; median $69.3 \rightarrow 60.0$), and one cell unsolvable at 1× is reached at 4×. The McNemar test is biased towards 4× by an unknown amount; the compute claim rests on the paired cold-start control. Sources: `runs/v1_long_train_subset_results.json`, `runs/v1_long_train_aggregate_15cell.json`.

S.2 Encoder-Geometry Diagnostic

Measurement	25×25 (d=64)	50×50 (d=128)
(A) Embedding geometry		
IsoScore (1 = isotropic)	0.044	0.034
Effective rank	20.1/64	25.6/128
Mean pairwise cosine	0.756	0.626
after mean-centring	0.021	0.019
Locality r_s, encoder†	0.124	0.057
Locality r_s, raw input features†	0.041	0.007
Locality r_s, *ground-truth coordinates*†	0.091	0.075
same probe, pure grid (E_{extra}=0)	0.978	0.980
(B) Retrieval gate, by embedding treatment		
Saturation ($\text{sim}^{\max} > 0.95$), raw	78.5 %	56.5 %
centred	12.3 %	9.2 %
whitened	0.05 %	0.43 %
Retrieval quality, raw	0.103	0.023
centred	0.102	0.025
whitened	0.090	0.022
Gate AUC, raw	0.584	0.636
(C) Metric probe: locality r_s by readout†		
Cosine (as deployed)	0.130	0.054
Learned ridge metric	0.224	0.154
Best learned metric (single best checkpoint)	0.287	0.195

Table S4: Encoder-geometry diagnostics for §4.6.1, 10 checkpoints per scale (five cells × warm/cold). **(A)** Geometry, including locality (Spearman r_s, embedding against hop distance). **(B)** Retrieval gate under four treatments (raw, mean-centred, whitened, percentile-calibrated): saturation, gate AUC, and retrieval quality (1.0 retrieves the nearest stored state, 0.0 a random one). **(C)** Metric probe: a learned metric (ridge, MLP) on frozen embeddings predicting hop distance. Warm against cold at 25×25: IsoScore 0.059 against 0.029, effective rank 23.7 against 16.5, mean cosine 0.666 against 0.845, warm better in all five scenarios; five pairs cannot reach significance (minimum Wilcoxon $p = 0.0625$), and none replicates at 50×50 (2/5, 3/5, 3/5 scenarios; $p \geq 0.31$). No claim is made that warm-starting mitigates encoder collapse. At 25×25 whitening cuts saturation from 78.5 % to 0.05 % (Wilcoxon $p = 0.002$) with retrieval quality unchanged ($p = 0.28$; centring $p = 0.92$). †Uninterpretable because shortcut edges compress hop distance (§3.2), so no conclusion is drawn. Sources: `runs/geometry_study_aggregate.json`, `runs/geometry_audit.json`, `runs/retrieval_gate.json`, `runs/metric_probe.json`; positive control via `scripts/analysis/locality_control.py` with $E_{\text{extra}} = 0$, values in `runs/thesis_support/item_c_locality_control.json`.

S.3 Cross-Distribution Transfer

Scenario	Grid	p_{deact}	V1 reach	V2 reach	V3 reach	V3 mean λ^{eff}
in-dist	100×100	0.30	2 / 10	3 / 10	2 / 10	0.321
new-seed	100×100	0.30	1 / 10	2 / 10	1 / 10	0.322
low-pert	100×100	0.15	2 / 10	2 / 10	3 / 10	0.312
high-pert	100×100	0.45	1 / 10	2 / 10	2 / 10	0.350
small-grid	50×50	0.30	1 / 10	1 / 10	2 / 10	0.337

Table S5: Cross-distribution transfer of V3 with the V1-trained agent and the V2 library, evaluated on five scenarios, ten queries per scenario (§4.5.2). V2 and V3, the expert library and the similarity gate are the superseded inference-time rules of Supplement S.4; inference-rule conditions are for cross-comparison only. Source: `runs/v3_transfer_benchmark.json`.

S.4 Superseded Inference-Augmentation Strand

Configuration	Agent	Library	λ rule	Goal-reach
V1 (baseline, no library)	V1-trained	—	—	11 / 25 (44 %)
V2-control (isolated mechanism effect)	V1-trained	yes	fixed 0.5	10 / 25 (40 %)
V2 published (agent + library, independent)	V2-trained	yes	fixed 0.5	8 / 25 (32 %)

Table S6: V2 on the 100×100 sweep. Superseded and retained for cross-comparison only; no claim in this thesis rests on the tables of this section. Two training-free inference-time mechanisms were layered over the trained V1 agent at 100×100. V2 blends a persistent expert library's recommendation at a fixed $\lambda_{\text{retr}} = 0.5$; V3 replaces the fixed coefficient with a similarity gate, $\lambda(s) = \lambda_{\max} \cdot \text{sim}^{\max}(s)$, where $\text{sim}^{\max}(s)$ is the highest cosine similarity between the current node's encoder embedding and any state embedding stored in the library, clipped below at zero, and λ^{eff} is the value of $\lambda(s)$ realised at a step, reported as a mean. V2-control uses the V1 checkpoint with the library at $\lambda_{\text{retr}} = 0.5$; V2 published uses a separately trained V2 agent. V1 and V2-control are at the matched 1,000-step budget (static regime of Table S10); V2 published comes from the 300-step headline sweep and is not budget-matched with them. Sources: `runs/unified_benchmark_100x100.json` (V1, static regime), `runs/sweep_v3_adaptive.json` (V2-control), `runs/sweep_v2_headline.json` (V2 published).

Configuration	Goal-reach	Mean cost (reached)	Mean cost ratio	Mean λ^{eff}
V1 (no library)	11 / 25 (44 %)	$8{,}934 \pm 4{,}858$	168.67	—
V2-control (fixed $\lambda = 0.5$)	10 / 25 (40 %)	$7{,}388 \pm 5{,}118$	132.16	0.50
V3 (adaptive λ, $\lambda_{\max} = 1.0$)	12 / 25 (48 %)	$8{,}097 \pm 4{,}092$	127.78	0.365

Table S7: V3 against V2-control and V1; one V1 checkpoint and the V2 library, inference rule varied, all at the matched 1,000-step budget. V2 reaches 10/25 and V3 12/25, results the static regime of Table S10 reproduces cell for cell; the 6/25 of the 300-step V1 sweep, on different evaluation states, is not a matched baseline. At the matched budget neither mechanism separates from V1 (b=1, c=2, $p = 1.0$ for V2; b=3, c=2, $p = 1.0$ for V3), nor does V3 from V2 (b=3, c=1, $p = 0.63$). Route quality is poor in all three: 132.2× Dijkstra for V2 and 127.8× for V3 against V1's 168.7×. Sources: `runs/sweep_v3_adaptive.json` (V2-control, V3), `runs/unified_benchmark_100x100.json` (V1, static regime).

$\lambda_{\max}$	Goal-reach	Mean cost (reached)	Mean λ^{eff}	Mean $\text{sim}^{\max}$
0.5	4 / 10	5,177	0.193	0.386
1.0	**6 / 10**	6,823	0.383	0.383
1.5	4 / 10	7,619	0.574	0.383
2.0	3 / 10	6,987	0.766	0.383

Table S8: V3 $\lambda_{\max}$ sensitivity on 10 cells (first two scenarios of each seed; not the first ten records of the sweep file). The $\lambda_{\max} = 1.0$ row is the headline run restricted to these cells, from `runs/sweep_v3_adaptive.json`; other rows from `runs/v3_lambda_ablation.json`.

Mode	Reach	Mean cost (reached)	Mean cost ratio	Mean λ^{eff}
V1 (no library)	6/10	5,344	84.0	—
V2 (fixed $\lambda_{\text{retr}} = 0.5$)	7/10	6,649	116.6	0.500
V3 (adaptive λ)	4/10	5,595	94.6	**0.974**

Table S9: V3 on the 10-cell long-train subset, one long-trained V1 agent per cell, library built at the evaluation state (matched encoder). McNemar V1 versus V3: $b = 2, c = 4, p = 0.6875$. The gate does not detect novelty: here it saturates at $\lambda^{\text{eff}} = 0.974$ and V3 reaches 4/10 cells against V1's 6/10, so the unsaturated gate of the headline sweep (mean $\lambda^{\text{eff}} = 0.365$) reflects encoder-state drift between library construction and evaluation; the representation itself is measured in §4.6.1. The evaluation uses a different perturbation realisation from Table S3, where the same agents reach all ten cells. Source: `runs/v3_long_train_subset_results.json`.

Regime (Δ)	Build	Goal-reach	Cost ratio	Latency (ms)	Survival
static	V1	11 / 25 (44 %)	168.7	541	—
	V2	10 / 25 (40 %)	132.2	5,962	—
	V3	**12 / 25 (48 %)**	127.8	5,011	—
	A*	24 / 25 (96 %)	1.36	21.4	—
	Cold	2 / 25 (8 %)	271.0	614	—
$\Delta = 50$	V1	11 / 25	165.3	612	1.00
	V2	8 / 25	125.1	5,955	1.00
	V3	11 / 25	130.1	6,227	1.00
	A**	24 / 25	1.36	27.3	n/a
	Cold	3 / 25	191.1	753	1.00
$\Delta = 30$	V1	11 / 25	208.8	629	1.00
	V2	10 / 25	123.5	6,047	1.00
	V3	10 / 25	138.9	6,696	1.00
	A**	24 / 25	1.36	20.9	n/a
	Cold	2 / 25	233.3	904	1.00
$\Delta = 15$	V1	13 / 25	161.7	763	1.00
	V2	9 / 25	101.2	7,474	1.00
	V3	11 / 25	190.2	7,682	1.00
	A**	24 / 25	1.36	15.8	n/a
	Cold	3 / 25	254.1	906	1.00

Table S10: Unified benchmark, $25 \times 5 \times 4 = 500$ evaluations (cells, conditions, regimes). Latency: mean per-trajectory wall-clock; survival: fraction of perturbation events after which the trajectory continued, 1 by construction (§3.7). *A* encounters zero perturbations in every regime (V1: 368–1,178) and its results are bit-identical across regimes, so its rows are only a reference for undisturbed path quality. Sources: `runs/unified_benchmark_100x100_aggregate.json`, `runs/unified_benchmark_100x100.json`.

S.5 Source Artefacts

Finding	Source artefact
25×25 sweep	`runs/sweep_phase3_final.json`
50×50 V1 sweep (1× tier, published June)	`runs/sweep_50x50/sweep_v1_50x50_1x.json`
Action-space A/B, reactive arm (50 × 50)	`runs/sweep_50x50_reactive_ctrl/`
Action-space A/B, masked arm (50 × 50)	`runs/sweep_50x50_masked_gpu/`
Action-space control, masked (100 × 100)	`runs/sweep_100x100_masked.json`
Masked multi-query control (50 × 50, 3 seeds)	`runs/fleet_fixed/reeval_seed*_50x50_mq_masked.json`

Finding	Source artefact
Per-run execution provenance (50×50 A/B arms only)	`provenance.json` (in each arm's run directory)
50×50 V1 sweep (4× tier)	`runs/sweep_50x50/sweep_v1_50x50_4x.json`
50×50 aggregate (gates, CI; McNemar stored asymptotic, exact values recomputed from per-cell files)	`runs/sweep_50x50/aggregate_50x50.json`
50×50 run record (config, S4-fix status)	`runs/sweep_50x50/REPORT.md`
Paired-test methods behind the gate scorecard	`runs/stats_reanalysis.json`
100×100 V1 sweep	`runs/sweep_v1_on_100x100.json`
100×100 V2 sweep + V2-control re-evaluation	`runs/sweep_v2_headline.json` (V2), `runs/sweep_v3_adaptive.json` (V2-control)
100×100 V3 sweep	`runs/sweep_v3_adaptive.json`
$\lambda_{\max}$ ablation	`runs/v3_lambda_ablation.json`
Cross-distribution transfer benchmark	`runs/v3_transfer_benchmark.json`
V1 long-train single cell	`runs/v1_long_train_results.json`
V1 long-train 15-cell subset (per-cell records)	`runs/v1_long_train_subset_results.json`
V1 long-train 15-cell aggregate (statistics, CI, McNemar)	`runs/v1_long_train_aggregate_15cell.json`
V3 long-train single-cell (legacy)	`runs/v3_long_train_results.json`
V3 long-train 10-cell subset (per-cell records)	`runs/v3_long_train_subset_results.json`
V3 long-train 10-cell aggregate (statistics)	`runs/v3_long_train_subset_aggregate.json`
Within-trajectory perturbation (25×25, conditions A/B/C)	`runs/v3_realtime_25x25_aggregate.json`
Unified comparison	`runs/unified_benchmark_100x100.json`, `runs/unified_benchmark_100x100_aggregate.json`
Unified sanity check	`runs/unified_benchmark_sanity.json`
Pareto plot data	`runs/unified_pareto_100x100.csv`
V1 perturbed-copies check (P1 gate)	`runs/perturbation_phase4.json`
Cold-start 4× control (per-cell + aggregate)	`runs/cold_4x_control_results.json`, `runs/cold_4x_control_aggregate.json`
Reward-design ablation (30 runs)	`runs/reward_ablation/reward_ablation_results.json`
Reward-shaping control (five arms, 25×25)	`runs/shaping_control/aggregate.json`, `runs/shaping_control/lambda_*.json`
Eval-state reachability audit	`runs/eval_reachability_audit.json`
Consolidated provenance inventory	`runs/INVENTORY.md`
Demonstration-source ablation (25×25), run log	`runs/demo_source_ablation.log`

Finding	Source artefact
Demonstration-source ablation, partial join (9/50)	`runs/demo_source_ablation_partial.json`
Per-arm demonstration diversity (25×25)	`runs/traces_25x25/*/*/seeding_diversity.json`
Sample efficiency (episodes-to-threshold, 25×25)	`runs/learning_curves_25x25.json`
Multi-query training, per seed (25×25)	`runs/fleet_fixed/fleet_results_seed*_25x25.json`
Multi-query held-out re-evaluation (150 queries)	`runs/fleet_fixed/reeval_seed*_25x25.json`
Goal-relative head, per seed	`runs/fleet_fixed/reeval_seed*_25x25_gr.json`
Fixed-code replication log (commit 4a231ba)	`runs/fleet_fixed/STATUS.txt`
Hindsight experience replay (5 seeds)	`runs/fleet_fixed/reeval_seed*_25x25_her.json`
Multi-query training, per seed (50 × 50)	`runs/fleet_fixed/fleet_results_seed*_50x50_mq.json`
Action-space A/B, reactive arm (25 × 25)	`runs/sweep_phase3_unmasked_ctrl.json`
Action-space A/B, masked arm (25 × 25)	`runs/sweep_phase3_masked.json`
Action-space A/B, both arms re-run on one recorded stack (25 × 25)	`runs/thesis_support/item_g_ab25_rerun/`
Termination-reason probe (reactive)	`runs/traces_reactive_probe/`
Masked replication, per-episode traces	`runs/traces_25x25_masked/`
Masked multi-query control (25 × 25, 5 seeds)	`runs/fleet_fixed/reeval_seed*_25x25_masked.json`
Encoder geometry, per checkpoint	`runs/geometry_audit.json`
Encoder geometry, aggregate (A/B/C studies)	`runs/geometry_study_aggregate.json`
Retrieval-gate treatments	`runs/retrieval_gate.json`
Learned-metric probe	`runs/metric_probe.json`
Pre-registered persistence ablation (50 × 50, masked)	`runs/persistence_ablation_50x50/aggregate.json`, `runs/persistence_ablation_50x50/cells_seed_*.json`
Persistence ablation pre-registration	`runs/persistence_ablation_50x50/PREREGISTRATION.md`
Demonstration-source ablation, traced full-pool and cold reference (25 × 25)	`runs/sweep_phase3_traced.json`
Multi-query held-out re-evaluation (50 × 50)	`runs/fleet_fixed/reeval_seed*_50x50_mq.json`
Exploration statistics on the 25 × 25 preset graph (regenerated)	`runs/thesis_support/item_a_preset_exploration_rules.json`

Finding	Source artefact
Within-trajectory Dijkstra-denominator inflation (regenerated)	`runs/thesis_support/item_b_realtime_dijkstra_inflation.json`
Locality positive control and hop IQR (regenerated)	`runs/thesis_support/item_c_locality_control.json`
Reward calibration and optimal returns (regenerated)	`runs/thesis_support/item_d_reward_calibration.json`
Exact graph diameters (regenerated)	`runs/thesis_support/item_e_exact_diameters.json`
Within-trajectory re-encode cost (regenerated)	`runs/thesis_support/item_f_reencode_timing.json`
Trainable parameter counts (regenerated)	`runs/thesis_support/item_h_param_counts.json`
Random-policy reach floor (regenerated)	`runs/thesis_support/item_i_random_policy_floor.json`
Pre-registered goal-adjacent seeding ablation (50×50, masked)	`runs/goaladj_ablation_50x50/aggregate.json`, `runs/goaladj_ablation_50x50/cells_seed_*.json`, `runs/goaladj_ablation_50x50/PREREGISTRATION.md`

Table S11: Source artefacts, relative to the project repository. Rows marked (regenerated) were produced in October 2026: three from the recorded analysis scripts, and the Dijkstra-inflation, exact-diameter and re-encode-timing artefacts from new deterministic analyses, all kept in `runs/thesis_support/scripts/`. Artefacts from two working trees and successive code revisions are consolidated in `runs/`, indexed by `runs/INVENTORY.md`, which records origin and superseded or incomplete status. Only artefacts produced after the protocol of §3.8.1 carry a `provenance.json` (`torch` build, device, compute capability, revision); earlier stacks cannot be established retrospectively. Superseded and not cited: `runs/sweep_50x50_masked/` (stack unrecorded). Incomplete: the demonstration-source ablation's aggregation aborted on a reproducibility guard, and `runs/demo_source_ablation_partial.json` holds only 9 seed-42 cells from a separate re-run, whose quantum-only costs differ from the logged run on three of four cells; the figures of §4.5.1 are recovered from the run log by `scripts/analysis/ablation_from_log.py`. Other artefacts in the repository are not evidence: the superseded 50×50 masked run (Table B.4); and perturbation-level diagnostics, a multiplier sweep on demonstration checkpoints (five cells per scale) and three density-robust retraining attempts (one 50×50 cell), outside the evaluation grid and not reported in Chapter 4. The reward-shaping control (§4.4.2, Table B.8) is the one declared exception to the provenance rule.